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(*This is for calculating the minimum of f(m) and g(m) for  $2\theta > s > t \geq 1$  and n even.
  m_{s,t} is the smallest nonnegative integer such that  $\text{Sqrt}[s-t-m]$  is an integer,
  m_f is the values of m such that f(m) attains its minimum,
  m_g is the values of m that g(m) attains its minimum.

  Note that we only list the case where the
  minimum of f(m) less or equal to the minimum of g(m).*)
(*****)
(*****)
Clear[s, t, m, a, b, i, j, n];
For[s = 2, s < 20, s++,
  For[t = 1, t < s, t++, f = 0; g = 0;
    For[m = 0, m < s - t + 1, m++, a = s * Sqrt[s - t - m] + m;
      b = s * Sqrt[s - t - m] + m + m - 2 Sqrt[s - t - m];
      If[IntegerQ[Sqrt[s - t - m]] && f < a, f = a; i = m];
      If[IntegerQ[Sqrt[s - t - m]] && g < b, g = b; j = m]
    ];
    If[f < g + 1,
      Print["s=", s, ", ", "t=", t,
        "; ", "min f(m)=",  $s * \frac{n}{2} - f$ , "; ", "min g(m)=",  $s * \frac{n}{2} - g$ ]];
    If[f < g + 1,
      For[m = 0, m < s - t + 1, m++,
        If[IntegerQ[Sqrt[s - t - m]], Print["m_{s,t}=", m]; Break[]]
      ];
      For[m = 0, m < s - t + 1, m++, a = s * Sqrt[s - t - m] + m;
        If[IntegerQ[Sqrt[s - t - m]] && SameQ[f, a], Print["m_f=", m]]
      ];
      For[m = 0, m < s - t + 1, m++, b = s * Sqrt[s - t - m] + m + m - 2 Sqrt[s - t - m];
        If[IntegerQ[Sqrt[s - t - m]] && SameQ[g, b], Print["m_g=", m]]
      ];
    ]
  ]
]
s=2, t=1; min f(m)=-2+n; min g(m)=-2+n
m_{s,t}=0
m_f=0
m_g=1
s=3, t=1; min f(m)=-4 +  $\frac{3n}{2}$ ; min g(m)=-4 +  $\frac{3n}{2}$ 
m_{s,t}=1
m_f=1
m_g=2
s=4, t=1; min f(m)=-6 + 2n; min g(m)=-6 + 2n
m_{s,t}=2

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$m_f=2$

$m_g=2$

$m_g=3$

$s=5, t=2; \min f(m)=-7+\frac{5n}{2}; \min g(m)=-7+\frac{5n}{2}$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$s=6, t=3; \min f(m)=-8+3n; \min g(m)=-8+3n$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$s=7, t=4; \min f(m)=-9+\frac{7n}{2}; \min g(m)=-9+\frac{7n}{2}$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$s=8, t=5; \min f(m)=-10+4n; \min g(m)=-10+4n$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$s=9, t=1; \min f(m)=-22+\frac{9n}{2}; \min g(m)=-22+\frac{9n}{2}$

$m_{\{s,t\}}=4$

$m_f=4$

$m_g=4$

$s=9, t=6; \min f(m)=-11+\frac{9n}{2}; \min g(m)=-11+\frac{9n}{2}$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$s=10, t=2; \min f(m)=-24+5n; \min g(m)=-24+5n$

$m_{\{s,t\}}=4$

$m_f=4$

$m_g=4$

$s=10, t=7; \min f(m)=-12+5n; \min g(m)=-12+5n$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$$s=11, t=3; \min f(m)=-26 + \frac{11n}{2}; \min g(m)=-26 + \frac{11n}{2}$$

$m_{\{s,t\}}=4$

$m_f=4$

$m_g=4$

$$s=11, t=8; \min f(m)=-13 + \frac{11n}{2}; \min g(m)=-13 + \frac{11n}{2}$$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$$s=12, t=4; \min f(m)=-28 + 6n; \min g(m)=-28 + 6n$$

$m_{\{s,t\}}=4$

$m_f=4$

$m_g=4$

$$s=12, t=9; \min f(m)=-14 + 6n; \min g(m)=-14 + 6n$$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$$s=13, t=5; \min f(m)=-30 + \frac{13n}{2}; \min g(m)=-30 + \frac{13n}{2}$$

$m_{\{s,t\}}=4$

$m_f=4$

$m_g=4$

$$s=13, t=10; \min f(m)=-15 + \frac{13n}{2}; \min g(m)=-15 + \frac{13n}{2}$$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$$s=14, t=6; \min f(m)=-32 + 7n; \min g(m)=-32 + 7n$$

$m_{\{s,t\}}=4$

$m_f=4$

$m_g=4$

$$s=14, t=11; \min f(m)=-16 + 7n; \min g(m)=-16 + 7n$$

$m_{\{s,t\}}=2$

$m_f=2$

$m_g=2$

$$s=15, t=7; \quad \min f(m) = -34 + \frac{15n}{2}; \quad \min g(m) = -34 + \frac{15n}{2}$$

$$m_{\{s,t\}}=4$$

$$m_f=4$$

$$m_g=4$$

$$s=15, t=12; \quad \min f(m) = -17 + \frac{15n}{2}; \quad \min g(m) = -17 + \frac{15n}{2}$$

$$m_{\{s,t\}}=2$$

$$m_f=2$$

$$m_g=2$$

$$s=16, t=1; \quad \min f(m) = -54 + 8n; \quad \min g(m) = -54 + 8n$$

$$m_{\{s,t\}}=6$$

$$m_f=6$$

$$m_g=6$$

$$s=16, t=8; \quad \min f(m) = -36 + 8n; \quad \min g(m) = -36 + 8n$$

$$m_{\{s,t\}}=4$$

$$m_f=4$$

$$m_g=4$$

$$s=16, t=13; \quad \min f(m) = -18 + 8n; \quad \min g(m) = -18 + 8n$$

$$m_{\{s,t\}}=2$$

$$m_f=2$$

$$m_g=2$$

$$s=17, t=2; \quad \min f(m) = -57 + \frac{17n}{2}; \quad \min g(m) = -57 + \frac{17n}{2}$$

$$m_{\{s,t\}}=6$$

$$m_f=6$$

$$m_g=6$$

$$s=17, t=9; \quad \min f(m) = -38 + \frac{17n}{2}; \quad \min g(m) = -38 + \frac{17n}{2}$$

$$m_{\{s,t\}}=4$$

$$m_f=4$$

$$m_g=4$$

$$s=17, t=14; \quad \min f(m) = -19 + \frac{17n}{2}; \quad \min g(m) = -19 + \frac{17n}{2}$$

$$m_{\{s,t\}}=2$$

$$m_f=2$$

$$m_g=2$$

$$s=18, t=3; \quad \min f(m) = -60 + 9n; \quad \min g(m) = -60 + 9n$$

$$m_{\{s,t\}}=6$$

$$m_f=6$$

$$m_g=6$$

$$s=18, t=10; \min f(m)=-40+9n; \min g(m)=-40+9n$$

$$m_{\{s,t\}}=4$$

$$m_f=4$$

$$m_g=4$$

$$s=18, t=15; \min f(m)=-20+9n; \min g(m)=-20+9n$$

$$m_{\{s,t\}}=2$$

$$m_f=2$$

$$m_g=2$$

$$s=19, t=4; \min f(m)=-63+\frac{19n}{2}; \min g(m)=-63+\frac{19n}{2}$$

$$m_{\{s,t\}}=6$$

$$m_f=6$$

$$m_g=6$$

$$s=19, t=11; \min f(m)=-42+\frac{19n}{2}; \min g(m)=-42+\frac{19n}{2}$$

$$m_{\{s,t\}}=4$$

$$m_f=4$$

$$m_g=4$$

$$s=19, t=16; \min f(m)=-21+\frac{19n}{2}; \min g(m)=-21+\frac{19n}{2}$$

$$m_{\{s,t\}}=2$$

$$m_f=2$$

$$m_g=2$$

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(*****)
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(*This is for calculating the minimum of f(m) and g(m) for  $2\theta > s > t \geq 1$  and n odd.
  m_{s,t} is the smallest nonnegative
  integer such that  $\text{Sqrt}[1+4(s-t-m)]$  is an integer,
  m_f is the values of m such that f(m) attains its minimum,
  m_g is the values of m that g(m) attains its minimum.
  Note that we only list the case where the
  minimum of f(m) less or equal to the minimum of g(m).*)
(*****)
(*****)
Clear[s, t, m, a, b, i, j];
For[s = 2, s < 2\theta, s++,
  For[t = 1, t < s, t++, f = 0; g = 0;
    For[m = 0, m < s - t + 1, m++, a =  $\frac{s}{2} * \text{Sqrt}[1 + 4(s - t - m)] + m$ ;
      b =  $\frac{s}{2} * \text{Sqrt}[1 + 4(s - t - m)] + m + m - \text{Sqrt}[1 + 4(s - t - m)]$ ;
      If[IntegerQ[Sqrt[1 + 4(s - t - m)]] && f < a, f = a; i = m];
      If[IntegerQ[Sqrt[1 + 4(s - t - m)]] && g < b, g = b; j = m]
    ];
    If[f < g + 1,
      Print["s=", s, ", ", "t=", t, "; ",
        "min f(m)=", s *  $\frac{n}{2} - f$ , "; ", "min g(m)=", s *  $\frac{n}{2} - g$ ]];
    If[f < g + 1,
      For[m = 0, m < s - t + 1, m++,
        If[IntegerQ[Sqrt[1 + 4(s - t - m)]]], Print["m_{s,t}=", m];
      Break[]];
    For[m = 0, m < s - t + 1, m++, a =  $\frac{s}{2} * \text{Sqrt}[1 + 4(s - t - m)] + m$ ;
      If[IntegerQ[Sqrt[1 + 4(s - t - m)]] && SameQ[f, a], Print["m_f=", m]]
    ];
    For[m = 0, m < s - t + 1, m++, b =  $\frac{s}{2} * \text{Sqrt}[1 + 4(s - t - m)] + m + m - \text{Sqrt}[1 + 4(s - t - m)]$ ;
      If[IntegerQ[Sqrt[1 + 4(s - t - m)]] && SameQ[g, b], Print["m_g=", m]]
    ];
  ];
]
]
s=2, t=1; min f(m)=-2+n; min g(m)=-2+n
m_{s,t}=1
m_f=1
m_g=1

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$$s=3, t=1; \quad \min f(m) = -\frac{9}{2} + \frac{3n}{2}; \quad \min g(m) = -\frac{9}{2} + \frac{3n}{2}$$

$$m_{\{s,t\}}=0$$

$$m_f=0$$

$$m_g=2$$

$$s=3, t=2; \quad \min f(m) = -\frac{5}{2} + \frac{3n}{2}; \quad \min g(m) = -\frac{5}{2} + \frac{3n}{2}$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=4, t=1; \quad \min f(m) = -7 + 2n; \quad \min g(m) = -7 + 2n$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=3$$

$$s=4, t=3; \quad \min f(m) = -3 + 2n; \quad \min g(m) = -3 + 2n$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=5, t=1; \quad \min f(m) = -\frac{19}{2} + \frac{5n}{2}; \quad \min g(m) = -\frac{19}{2} + \frac{5n}{2}$$

$$m_{\{s,t\}}=2$$

$$m_f=2$$

$$m_g=4$$

$$s=5, t=4; \quad \min f(m) = -\frac{7}{2} + \frac{5n}{2}; \quad \min g(m) = -\frac{7}{2} + \frac{5n}{2}$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=6, t=1; \quad \min f(m) = -12 + 3n; \quad \min g(m) = -12 + 3n$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$m_g=5$$

$$s=6, t=5; \quad \min f(m) = -4 + 3n; \quad \min g(m) = -4 + 3n$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=7, t=2; \quad \min f(m) = -\frac{27}{2} + \frac{7n}{2}; \quad \min g(m) = -\frac{27}{2} + \frac{7n}{2}$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=7, t=6; \quad \min f(m) = -\frac{9}{2} + \frac{7n}{2}; \quad \min g(m) = -\frac{9}{2} + \frac{7n}{2}$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=8, t=3; \quad \min f(m) = -15 + 4n; \quad \min g(m) = -15 + 4n$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=8, t=7; \quad \min f(m) = -5 + 4n; \quad \min g(m) = -5 + 4n$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=9, t=4; \quad \min f(m) = -\frac{33}{2} + \frac{9n}{2}; \quad \min g(m) = -\frac{33}{2} + \frac{9n}{2}$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=9, t=8; \quad \min f(m) = -\frac{11}{2} + \frac{9n}{2}; \quad \min g(m) = -\frac{11}{2} + \frac{9n}{2}$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=10, t=5; \quad \min f(m) = -18 + 5n; \quad \min g(m) = -18 + 5n$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=10, t=9; \quad \min f(m) = -6 + 5n; \quad \min g(m) = -6 + 5n$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=11, t=6; \quad \min f(m) = -\frac{39}{2} + \frac{11n}{2}; \quad \min g(m) = -\frac{39}{2} + \frac{11n}{2}$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=11, t=10; \min f(m)=-\frac{13}{2}+\frac{11n}{2}; \min g(m)=-\frac{13}{2}+\frac{11n}{2}$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=12, t=1; \min f(m)=-35+6n; \min g(m)=-35+6n$$

$$m_{\{s,t\}}=5$$

$$m_f=5$$

$$m_g=5$$

$$s=12, t=7; \min f(m)=-21+6n; \min g(m)=-21+6n$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=12, t=11; \min f(m)=-7+6n; \min g(m)=-7+6n$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=13, t=2; \min f(m)=-\frac{75}{2}+\frac{13n}{2}; \min g(m)=-\frac{75}{2}+\frac{13n}{2}$$

$$m_{\{s,t\}}=5$$

$$m_f=5$$

$$m_g=5$$

$$s=13, t=8; \min f(m)=-\frac{45}{2}+\frac{13n}{2}; \min g(m)=-\frac{45}{2}+\frac{13n}{2}$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=13, t=12; \min f(m)=-\frac{15}{2}+\frac{13n}{2}; \min g(m)=-\frac{15}{2}+\frac{13n}{2}$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=14, t=3; \min f(m)=-40+7n; \min g(m)=-40+7n$$

$$m_{\{s,t\}}=5$$

$$m_f=5$$

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m_g=5
s=14, t=9; min f(m)=-24+7 n; min g(m)=-24+7 n
m_{s,t}=3
m_f=3
m_g=3
s=14, t=13; min f(m)=-8+7 n; min g(m)=-8+7 n
m_{s,t}=1
m_f=1
m_g=1
s=15, t=4; min f(m)=-\frac{85}{2}+\frac{15 n}{2}; min g(m)=-\frac{85}{2}+\frac{15 n}{2}
m_{s,t}=5
m_f=5
m_g=5
s=15, t=10; min f(m)=-\frac{51}{2}+\frac{15 n}{2}; min g(m)=-\frac{51}{2}+\frac{15 n}{2}
m_{s,t}=3
m_f=3
m_g=3
s=15, t=14; min f(m)=-\frac{17}{2}+\frac{15 n}{2}; min g(m)=-\frac{17}{2}+\frac{15 n}{2}
m_{s,t}=1
m_f=1
m_g=1
s=16, t=5; min f(m)=-45+8 n; min g(m)=-45+8 n
m_{s,t}=5
m_f=5
m_g=5
s=16, t=11; min f(m)=-27+8 n; min g(m)=-27+8 n
m_{s,t}=3
m_f=3
m_g=3
s=16, t=15; min f(m)=-9+8 n; min g(m)=-9+8 n
m_{s,t}=1
m_f=1
m_g=1
s=17, t=6; min f(m)=-\frac{95}{2}+\frac{17 n}{2}; min g(m)=-\frac{95}{2}+\frac{17 n}{2}

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$$m_{\{s,t\}}=5$$

$$m_f=5$$

$$m_g=5$$

$$s=17, t=12; \min f(m)=-\frac{57}{2}+\frac{17n}{2}; \min g(m)=-\frac{57}{2}+\frac{17n}{2}$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=17, t=16; \min f(m)=-\frac{19}{2}+\frac{17n}{2}; \min g(m)=-\frac{19}{2}+\frac{17n}{2}$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=18, t=7; \min f(m)=-50+9n; \min g(m)=-50+9n$$

$$m_{\{s,t\}}=5$$

$$m_f=5$$

$$m_g=5$$

$$s=18, t=13; \min f(m)=-30+9n; \min g(m)=-30+9n$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=18, t=17; \min f(m)=-10+9n; \min g(m)=-10+9n$$

$$m_{\{s,t\}}=1$$

$$m_f=1$$

$$m_g=1$$

$$s=19, t=8; \min f(m)=-\frac{105}{2}+\frac{19n}{2}; \min g(m)=-\frac{105}{2}+\frac{19n}{2}$$

$$m_{\{s,t\}}=5$$

$$m_f=5$$

$$m_g=5$$

$$s=19, t=14; \min f(m)=-\frac{63}{2}+\frac{19n}{2}; \min g(m)=-\frac{63}{2}+\frac{19n}{2}$$

$$m_{\{s,t\}}=3$$

$$m_f=3$$

$$m_g=3$$

$$s=19, t=18; \min f(m)=-\frac{21}{2}+\frac{19n}{2}; \min g(m)=-\frac{21}{2}+\frac{19n}{2}$$

$$m_{\{s,t\}}=1$$

```
m_f=1
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m_g=1
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